

Top Selling Vendors
Show the top 5 vendors who sold the highest total quantity of products across all market dates.

Show vendor id, vendor name and total quantity sold.

select * from customer_purchases; -- vendor id is part of it [4221]

select * from vendor; -- vendor id [9 vendors]

-- inner join - I want vendors who have sold products

select

 v.vendor_id,

 v.vendor_name,

 -- to get total quantity I need to sum up the quantities

sum(cp.quantity) **as** total_quantity

from vendor v

inner join customer_purchases cp

on v.vendor_id = cp.vendor_id

-- because I perform sum aggregation

group by v.vendor_id, v.vendor_name

order by total_quantity desc

limit 5;

Booths That Were Never Assigned

Find booths from the booth table that were never assigned to any vendor.

show booth_number, booth_price_level, booth_description, booth_type

select * from booth; -- all available booths are present here

select * from vendor_booth_assignments; -- who are all of the vendors using different booth on different market date

select

 b.booth_number,

 b.booth_price_level,

 b.booth_description,

 b.booth_type

from booth b

left join vendor_booth_assignments vba

on b.booth_number = vba.booth_number

where vba.vendor_id is null;

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# Repeat Customers
# List all customers who have made purchases on at least 3 different market dates.
# Show their first name, last name, zip code, and count of distinct market dates.
# sorted by first name
select * from customer limit 10;
select * from customer_purchases limit 10;
-- inner join
select
    c.customer_first_name, -- 1
    c.customer_last_name, -- 2
    c.customer_zip,
    count(distinct cp.market_date) as market_dates
from customer c
inner join customer_purchases cp
on c.customer_id = cp.customer_id
group by c.customer_first_name, c.customer_last_name, c.customer_zip
having market_dates > 3
order by c.customer_first_name;

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# Average Spend per Customer per Visit
# show customer_id, average_amount_spent_per_visit
-- This is a very vague question
select * from customer_purchases;
-- market_date, customer_id, quantity, cost_per_qty
with sum_of_purchase_in_each_visit as (
    select
        customer_id,
        market_date,
        round(sum(quantity * cost_to_customer_per_qty), 2) as
total_cost_in_visit
    from customer_purchases
    group by customer_id, market_date
)
select customer_id, avg(total_cost_in_visit) from sum_of_purchase_in_each_visit
group by customer_id;
-- sum / total_visit

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Product Price Variations

For each product, show the minimum and maximum price charged to customers (cost_to_customer_per_qty).

Sort by the difference in price (highest difference first).

show the product_id, product_name, min_price, max_price, price_diff

select * from product;

select * from customer_purchases;

Understocked Products

List all products whose inventory (in vendor_inventory) has ever gone below 10 units.

Include the product name, market date, vendor name, and quantity.

select * from vendor_inventory;

select * from vendor;

select * from product;

Booth Assignment Changes

Find vendors who have changed booths across different market dates.

Show vendor id, vendor name, number of unique booths they've been assigned to, and total market appearances.

select * from vendor;

select * from vendor_booth_assignments;

Busiest Market Days

Which 3 market dates had the highest number of total transactions (rows in customer_purchases)?

Show market_date, market_day, market_season, market_rain_flag, market_snow_flag,

highest_num_of_tot_transactions

select * from market_date_info;

select * from customer_purchases;

First Purchase per Customer

Find the first-ever purchase made by each customer.

show customer_id, market_date, product_id

select customer_id, market_date, quantity, cost_to_customer_per_qty from customer_purchases order by customer_id, market_date;

Top 3 Booths by Usage Count

Count how often each booth has been assigned and rank the most used booths.

select * from vendor_booth_assignments;

Detect Price Jumps for a Product

Find when the price for a product changed from one market date to the next.

Show product_id, market_date, cost_to_customer_per_qty, next_price, next_date

select * from customer_purchases;

Compare Current and Previous Day's Inventory for Vendors

Find changes in quantity held by vendors day-to-day.

show vendor_id, product_id, market_date, quantity, prev_quantity

select * from vendor_inventory;

Top Selling Products per Category

For each product category, rank products based on total quantity sold

Show: category_name, product_name, total_quantity, rank_in_category

select * from customer_purchases;

select * from product;

select * from product_category;